



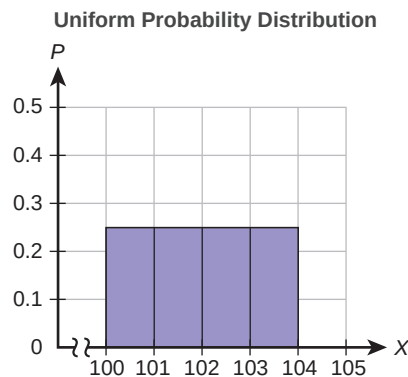
Continuous Random Variables

Areas within probability distribution graphs can be used to compute probability.

Exploring Continuous Random Variables

An incubator is being used to hatch chickens. The incubator temperatures X , in degrees Fahrenheit, are distributed according to this probability distribution.

When all values of a random variable X are equally likely to occur, the resulting probability distribution is called a **uniform probability distribution**.



For the incubator, all of the values of X , from 100°F to 104°F are equally likely to occur. Because any temperature between these values, such as 101.5°F or 103.9°F, may occur, X , in this case, is considered to be a continuous random variable.

Definition

A variable X is a **continuous random variable** if it can take on any value in the possible range of the random variable.

► Q&A

Q How is a discrete random variable different from a continuous random variable?

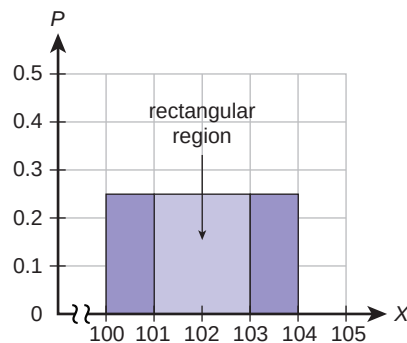
A A discrete random variable has a limited number of values, and a continuous random variable has an infinite number of values.

Using Area to Determine Probability

When X is a continuous random variable, the resulting probability distribution graph is a geometric shape with a total area of 1. The probability that $a \leq X \leq b$ will be the area of the shape bordered by the values a and b .

For instance, suppose the temperature of the incubator is taken at 10 a.m. What is the probability that the temperature is between 101°F and 103°F?

To answer this question, you can compute the area of the rectangular region where $101 \leq X \leq 103$, which is shown here.



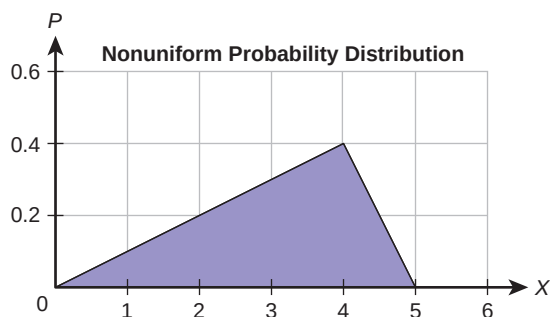
The area of the rectangular region is the product of the base and the height.

$$2 \cdot 0.25 = 0.5$$

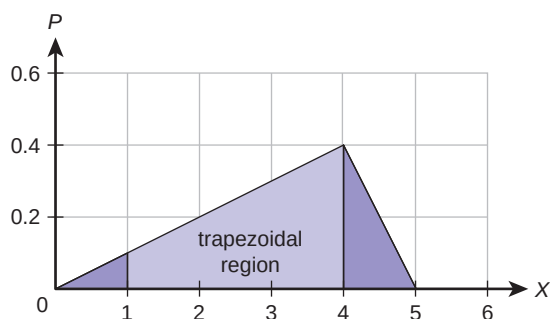
There is a 50% probability that the temperature will be between 101°F and 103°F at 10 a.m.

Using Continuous Nonuniform Probability Distributions

Here is another continuous probability distribution that is not uniform and is in the shape of a triangle.



Suppose we want to know $P(1 \leq X \leq 4)$, which is the probability that X is between 1 and 4.



The region bounded by $X = 1$ and $X = 4$ is a trapezoid with base lengths 0.1 and 0.4. The height of the trapezoid is 3.

$$\begin{aligned}\text{area} &= \frac{1}{2}h (\text{base}_1 + \text{base}_2) \\ &= \frac{1}{2} \cdot 3 (0.1 + 0.4) \\ &= 0.75\end{aligned}$$

The probability that X is between 3 and 4 is 0.75, or 75%.